Cover Page

Problem Statement / Concept Title	**Smart Food Waste Management and Menu Optimization System**
Theme	*[Theme as selected on the portal]*
Team Name (Registered on portal)	*[Registered team name]*
Team ID (Registered on portal)	*[Registered team ID]*
State / UT	*[State / UT]*

Team Name

Problem – Proposed Solution

Core problem: Meals in old-age homes, PG hostels and community messes are cooked on rough estimates, not actual demand – so edible food is thrown away.

Proposed Solution

- Load cell + HX711 weighs discarded food
- Ultrasonic sensor tracks bin fill level
- ESP32 sends data over Wi-Fi to cloud
- Python finds patterns; dashboard shows reports and alerts

How It Addresses the Problem

- Real-time measurement replaces guesswork
- Reveals high-waste items and peak timings
- Recommends food quantity per meal
- Menu changes cut over-preparation and cost

Innovation & Uniqueness

- Closed loop: measure waste, then recommend menu
- Dual sensing: weight + fill level
- Low-cost build (est. ₹12,000–16,000)
- Roadmap: AI-based demand prediction

Team Name

Technical Approach

Hack for Social Cause

1 SENSE
Load cell + HX711, ultrasonic

→

2 COLLECT
ESP32 microcontroller

→

3 STORE
Firebase / MySQL over Wi-Fi

→

4 ANALYSE
Python trend analysis

→

5 MONITOR
Dashboard, reports, alerts

→

6 OPTIMISE
Menu and quantity advice

Technologies

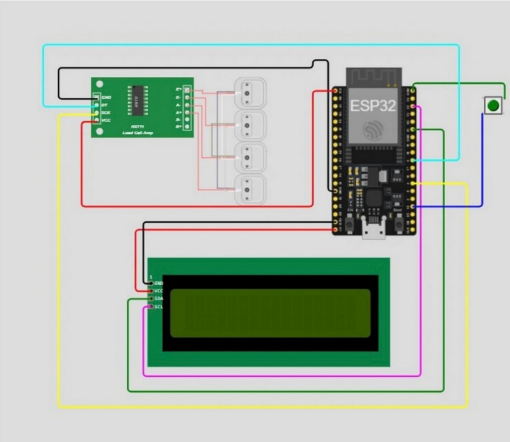
ESP32, load cell + HX711, ultrasonic sensor, LCD
Firebase / MySQL, Python, web dashboard

GitHub / GitLab

[Add repository link]

Demo video

[Add demo video link]



Circuit design: ESP32, HX711 + load cells, LCD



Dashboard prototype

Extent of AI use

Python trend analysis now; ML demand forecasting planned

Team Name

Impact Potential

Impact on target audience

- Old-age homes, PG hostels, college messes and community kitchens cook what is actually needed
- Residents get menus planned around what they eat
- Staff and wardens decide with live data, not guesswork
- Less food and fewer funds wasted day to day

Benefits

Social

Donated food is used well

Economic

Lower ingredient and running costs

Environmental

Less organic waste to dispose

Stakeholders

- Residents & students
- Institution management
- Mess staff & wardens
- Donors & NGOs

Team Name

Feasibility and Risk Analysis

Feasibility

₹12–16K

estimated total build cost

- Low-cost, widely available parts
- Standard stack: ESP32, Firebase, Python
- Circuit designed, dashboard prototyped

Challenges & Strategies

Risk	Strategy
Wi-Fi drops	Buffer readings on ESP32; resend when online
Sensor drift, wet or uneven waste	Regular calibration; sturdy mounting frame
Staff adoption	Simple dashboard; live weight on bin LCD
Waste not linked to a dish	Log waste per meal session to match menu
Cloud running cost	Low-cost cloud tier (budget ₹1,000–1,500)

Team Name

Team

Team Composition

Swetha M
Team Lead

Sri Moksha M
Team member – [add role]

Jayashree D M
Team member – [add role]

Ms. S. Gowrishwari
Mentor

Ms. T. Mangayarkarasi
Principal Investigator

Challenges

- *[Add the challenges your team faced]*
- *[e.g. sensor calibration, data transfer]*

Our Journey

- *[Add anything else you want to share]*

Team Name

References

ESP32 documentation (Espressif)
https://docs.espressif.com

HX711 load cell amplifier datasheet (Avia Semiconductor)
https://cdn.sparkfun.com/datasheets/Sensors/ForceFlex/hx711_english.pdf

Firebase documentation (Google)
https://firebase.google.com/docs

pandas documentation – Python data analysis
https://pandas.pydata.org/docs/

[Add any other papers, reports or surveys your team referred to]